

Time: Three Hours

Maximum Marks: 75

SECTION A – (10 × 2 = 20 marks)**Answer any TEN questions**

1. Write down the equation of $(1 - x)^2$.
2. Write down the value of $\left(\frac{e^x + e^{-x}}{2}\right)$.
3. Define Hermitian Matrix.
4. State Cauchy – Hamilton theorem.
5. Prove that $\Delta = E - 1$.
6. Prove that $\nabla = 1 - E^{-1}$.
7. State Lagrange Interpolation formula.
8. Write down the expansion of $\sin\theta$.
9. Prove that $\cosh^2 x = \sinh^2 x + 1$.
10. If $x = r \cos \theta$; $y = r \sin \theta$, then find $\frac{\partial(x,y)}{\partial(r,\theta)}$.
11. State Leibnitz's theorem.
12. Find the n^{th} derivative of $\text{Log}(ax + b)$.

SECTION B – (5 × 5 = 25 marks)**Answer any FIVE questions**

13. Prove that $\frac{e-1}{e+1} = \frac{\frac{1}{2!} + \frac{1}{4!} + \frac{1}{6!} + \dots}{\frac{1}{1!} + \frac{1}{3!} + \frac{1}{5!} + \dots}$.
14. Find the sum to infinity of the series $1 + \frac{3}{4} + \frac{3.5}{4.8} + \frac{3.5.7}{4.8.12} + \dots$.
15. Show that the matrix $A = \begin{pmatrix} -2/3 & 1/3 & 2/3 \\ 2/3 & 2/3 & 1/3 \\ 1/3 & -2/3 & 2/3 \end{pmatrix}$ is orthogonal.
16. Find the Characteristics equation of $A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$.
17. Using Lagrange's Interpolation Formula find y (10)

$x :$	5	6	9	11
$y :$	12	13	14	16
18. Expand $\tan 5\theta$ in terms of $\tan \theta$.
19. Find the maximum and minimum value of $x^2 - xy + y^2 - 2x + y$.

SECTION C – (3 × 10 = 30 marks)

Answer any THREE questions

20. Show that $\log \sqrt{12} = 1 + \left(\frac{1}{2} + \frac{1}{3}\right)\frac{1}{4} + \left(\frac{1}{4} + \frac{1}{5}\right)\frac{1}{4^2} + \left(\frac{1}{6} + \frac{1}{7}\right)\frac{1}{4^3} + \dots$

21. Verify Cauchy Hamilton theorem for the matrix $A = \begin{pmatrix} 3 & 1 & 2 \\ 4 & 3 & 1 \\ 2 & 5 & 4 \end{pmatrix}$

22. Prove that $\cos^4 \theta \sin^2 \theta = \frac{-1}{32} [\cos 6\theta + 2\cos 4\theta - \cos 2\theta - 2]$.

23. Using Newton's forward interpolation formula find y at $x = 2.5$ from the table.

$x :$	0	1	2	3	4
$y :$	7	10	13	22	43

24. If $y = \sin^{-1} x$, Prove that $(1 - x^2) y_{n+2} - (2n + 1)xy_{n+1} -$

$$n^2 y_n = 0.$$